Adjacent angles on one page

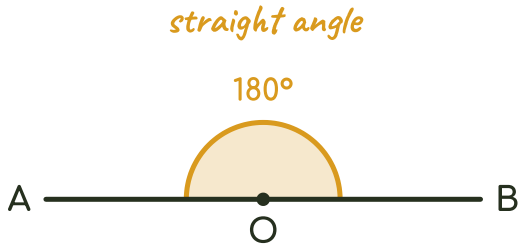


Hi! Here is everything important about adjacent angles: what they are, why together they make 180°, and how to solve problems with bisectors and ratios.

A straight angle

1

Rays OA and OB are **opposite rays**: together they make a straight line. The angle between them is **180°**.

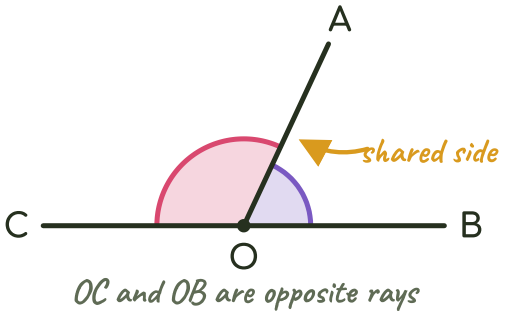


That's half a full turn (360°).

Adjacent angles

2

Two angles are **adjacent on a straight line** if they share one side and their other sides are opposite rays.



∠COA and ∠AOB are adjacent: together they make the straight angle COB.



Adjacent angles are two pieces of one straight angle.

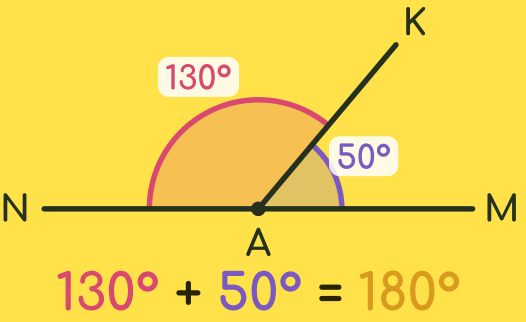
3

Key rule: adjacent angles add up to 180°

∠NAK + ∠KAM = 180°. However you tilt ray AK, the two adjacent angles always make a straight angle together.



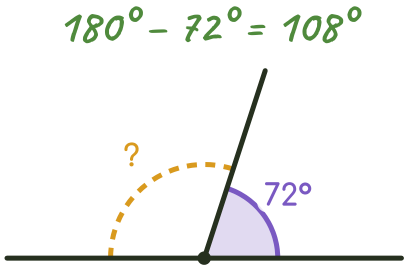
One angle grows, its neighbour shrinks. Together they always make 180°!



Find the partner

4

Know one angle? Subtract it from 180°: **partner = 180° − known angle**.

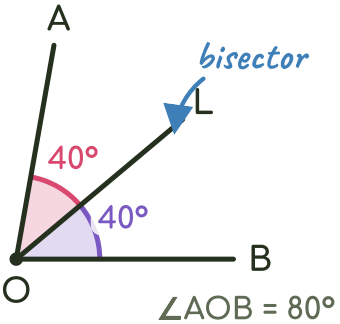


Check: 72° + 108° = 180°.

The angle bisector

5

The **bisector** is the ray that splits an angle in half. Angle = 2 × half.





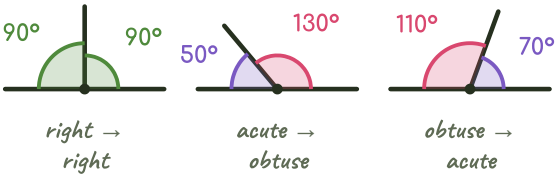
Split it fairly, like a treat for two!

Three consequences

6

The partner of a **right** angle is right. The partner of an **acute** angle is obtuse. The partner of an **obtuse** angle is acute.

Also: if two angles are equal, their partners are equal too. Two angles that are both partners of the same angle are equal.



right → right

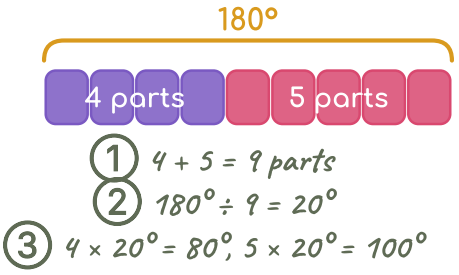
acute → obtuse

obtuse → acute

Angles in the ratio 4 : 5

7

Two adjacent angles are in the ratio 4 : 5. Split 180° into **parts**.

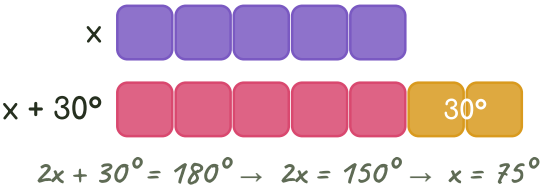


Check: 80° + 100° = 180°, and 80 : 100 = 4 : 5.

One is 30° bigger

8

Call the smaller angle **x** and the bigger one **x + 30°**. Together they make 180°.



$2x + 30° = 180° \rightarrow 2x = 150° \rightarrow x = 75°$

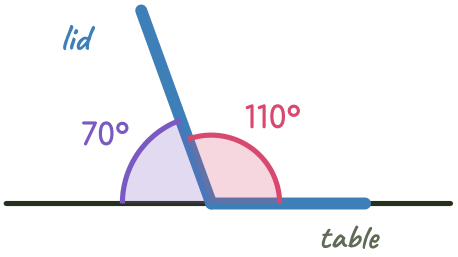
Answer: 75° and 105°. Check: 105° − 75° = 30°.

Angles in real life

9

A laptop lid is opened to 110° from the keyboard. So it makes **70°** with the rest of the table line.

Builders, navigators and game designers work out angles like these every day.





Check yourself: add the two adjacent angles you found. Do you get 180°? Then you're right!

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Built on his slides and lessons. Made by Mykola Stepanov for his daughter Varia